

ASTEROID FLYBYS FROM THE L2 SUN-EARTH LAGRANGE POINT

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Abstract

Over a mission lifespan, there are several phases where a spacecraft may underspend its fuel budget, leaving a limited but non-negligible amount of propellant at the end of the nominal mission timeline, unlocking possibilities to perform an add-on mission with a profile contrastive to the nominal. The present research was prompted by example of the ESA GAIA mission, where the remaining propellant generated discussion for such secondary science. One option explores Near-Earth Object (NEO) encounters from the Sun-Earth L2 (SEL2) Lagrange Point within a 2-year departure window spanning 01/01/25 – 01/01/27. Mission requirements including limitations on total delta-v (up to 150ms^{-1}), Time of Flight (TOF, up to 9 months), and relative flyby velocities (up to 5kms^{-1}) acted as constraints to the trajectory designs. Two Mission Scenarios (MS) were assessed and utilised escapes along the SEL2 unstable manifolds. MS 1 employed an exterior transfer towards the first Aphelion point, whilst MS 2 considered an interior escape from SEL2, followed by an Earth-powered flyby before asteroid rendezvous. NEO filtering evaluated target asteroid reachability by assessing the Minimum Orbital Intersection Distance (MOID) between the spacecraft and the NEO against a threshold value calculated from delta-v mission constraints. Desirable traits such as the Absolute Magnitude were also considered. Across the two designs, 91 preliminary candidates were deemed potentially reachable from asteroid filtering - Pork chop plots were built to provide reliable starting guesses for the candidate flybys and pave the way for trajectory optimisation. The latter was achieved by employing a single shooting method with mission constraints cost functions. The 10 most desirable candidates from MS 1 and all 20 from MS 2 underwent the optimisation scheme. Of the 30 candidates evaluated, 14 asteroids met at least one mission constraint, 11 meeting the delta-v constraint. Of these 11, 9 had relative flyby velocities below 5kms^{-1} . 2 NEOs from MS 2 returned feasible trajectories below 11 months, the closest to the TOF requirement. Generally, MS 1 returned a higher candidate yield but suffered in its TOF in comparison to MS 2. MS 2 had less than half the manifold transfer duration to MS 1 but provided difficulty in optimising, as the initial guess was not deterministic unlike MS 1. From just the initial batches, so far it was demonstrated that not only can asteroid flybys be achieved within this delta-v, but also they can be done for significantly less than 150ms^{-1} if desired.

Keywords: Trajectory Design, Asteroid Missions, Three Body Problem, Low Thrust, Invariant Manifolds, Planetary Flybys

1. Introduction

Once placed at SEL2 a spacecraft is able to remain there with only occasional station keeping, not by orbiting any particular body, but rather being subject to a gravitational tug-of-war between celestial bodies. Coupling this with excess margins often originating from launcher dispersion stages, leaves leftover fuel potentially wasted. With an increasing number of missions deployed into operational orbit into SEL2, there are growing opportunities to complete secondary science missions such as asteroid flybys after accomplishing the primary mission. To locate SEL2, it is necessary to exploit the CR3BP domain. The system also opens the window up to unstable manifolds, analogous to sea currents in space, allowing a satellite to be carried along in a drift to design dependent desired locations for next to no fuel cost.

From a scientific perspective, asteroids are remnants dating back to the early formation of our solar system, and their orbital motions allow us to treat them as ‘tracer particles’, acting as flying history recordings of how major celestial bodies have moved over time, akin to Earth’s tectonic plates. From a planetary defence perspective, there is an unnerving awareness of the potentially catastrophic extinction-level events imposed if one large enough impacts the Earth’s surface. In the present, there is even the ambitious and economically revolutionary goal of asteroid mining, since the majority of them have never melted and undergone geochemical differentiation. This means that many asteroids are made up of a fine-grained mixture of all the precious materials that exist within Earth’s core, mantle, and crust, but homogeneously present throughout their bodies [1].

The majority of interplanetary missions during the 20th century focused on major planetary exploration, but this was followed by a surge on interest in the investigation of minor bodies [2]. An interest is understatement, but rather a conquest of new firsts for space exploration [3]:

- NASA's NEAR (Near Earth Asteroid Rendezvous) Shoemaker, first comprehensive mapping and landing on 433 Eros in 2001.
- JAXA's Hayabusa, first successful sample return mission to Itokawa in 2003 (followed by Hayabusa 2 to Ryugu in 2014).
- NASA's DART (Double Asteroid Redirection Test), first technological demonstration to activity redirect a binary asteroid system (Didymos - Dimorphos) in 2021.

In the past, it has also been raised that the scenarios presented in this study parallel that of the Comet Interceptor, a piggyback mission aboard ARIEL with a total capacity of 650ms^{-1} . Each of the aforementioned missions and the Comet Interceptor included, have had or have their eyes set on minor bodies as a primary mission or secondary mission statement – *but never as a consideration in the event of leftover fuel*.

This paper provides trajectory design and optimisation for flybys to NEOs from SEL2 under low-cost mission constraints laid out by previous industrial collaboration. A hard limit on delta- v of 150ms^{-1} or lower was essential. Soft constraints included a TOF of around 9 months or lower, and a relative approach velocity of 5kms^{-1} or lower. Additional desirable parameters included the scientific value of the asteroid (i.e., the diameter) in question. Ephemerides tracking tools such as the JPL Horizons Small Bodies Database [4] have over 35,0000 documented NEOs, hence comprehensive trade-off studies must also be undergone to determine most appropriate candidate asteroids to target in the mission design process. The overarching aim is to reduce the full asteroid database significantly by filtering on the MOID, and then the most applicable asteroids will act as the final position vectors for the spacecraft to reach within the single shooting method trajectory optimisation.

2 low-cost scenarios to NEOs are presented in this paper. Both exploit invariant manifolds in the CR3BP domain to escape SEL2 to the Aphelion in MS 1 (see Figure 1) and to the Perigee in MS 2 (see Figure 2). From these locations, Patched Conics approximations are applied thereafter in the 2BP domain to rendezvous with final candidates: a deterministic single burn at the Aphelion, and an Earth-Powered Flyby (EPF) at the Perigee.

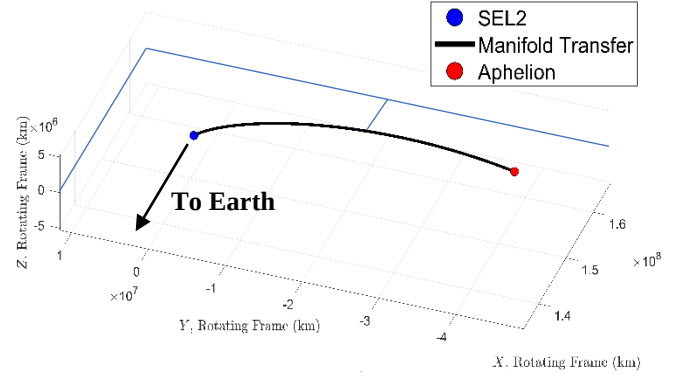


Figure 1. MS 1 SEL2 escape in in the Synodic frame.

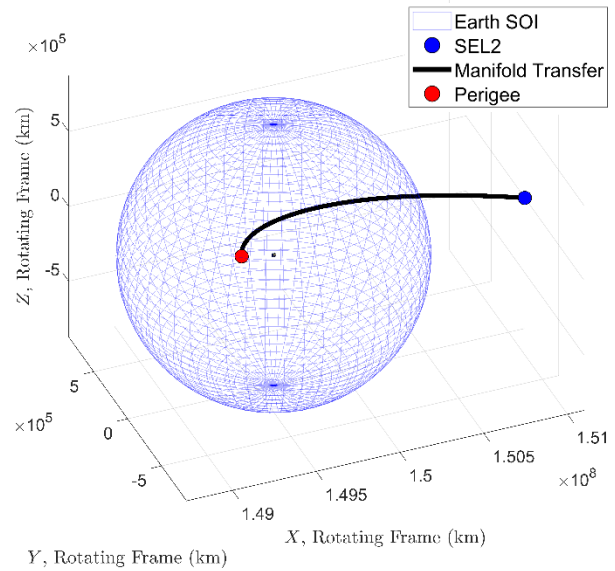


Figure 2. MS 2 SEL2 Escape and Earth Powered Flyby to Earth's Sphere of Influence boundary in the Synodic frame.

2. Circular Restricted Three Body Problem

The CR3BP is by no means a new field of study; for equilibrium points, the purposes of this study, concise introductions to Libration point theory by McInnes [6] were used. The key assumptions to bring about the CR3BP from its trickier predecessor are as follows: all three bodies are assumed point masses. Suppose also that two massive spherical bodies $m_{\odot}$ (the Sun) and $m_{\oplus}$ (the Earth) - referred to as the 'Primaries' - are in circular, Keplerian motion, whilst the spacecraft is moving among them, with relatively infinitesimal mass and unable to affect the Primaries [6].

2.1 Equations of Motion

To formulate a mathematical expression describing the motion of the spacecraft in the Sun-Earth system, two reference frames are introduced (see Figure 3). Let $P = \{B | \hat{e}_e, \hat{e}_p, \hat{e}_h\}$ be the Perifocal (inertial) frame and let $S = \{B | \hat{e}_r, \hat{e}_\theta, \hat{e}_h\}$ be the Synodic (rotating) frame,

centred in the barycentre B of the Primaries [6]. Next a crucial variable of the CR3BP - the Mass-Ratio Parameter, $\bar{\mu}$:

$$\bar{\mu} = \frac{m_{\oplus}}{m_{\odot} + m_{\oplus}} \quad (1)$$

Alongside $\bar{\mu}$, the Equations of Motion (EOM) can also be simplified by way of Normalising Units (NU):

$$1LU \text{ (Length Unit)} = a \quad (2)$$

$$1TU \text{ (Time Unit)} = \frac{1}{n} \quad (3)$$

$$1MU \text{ (Mass Unit)} = (m_1 + m_2) \quad (4)$$

Derived from the Mean Motion n and Semi-Major Axis a . Substituting both $\bar{\mu}$ and NU returns the common expression of the CR3BP EOM:

$$\begin{cases} \ddot{x} - 2\dot{y} = \Omega_x \\ \ddot{y} - 2\dot{x} = \Omega_y \\ \ddot{z} = \Omega_z \end{cases} \quad (5)$$

Where Ω is the Effective Potential:

$$\Omega = \frac{1}{2}(x^2 + y^2) + \frac{1 - \bar{\mu}}{r_1} + \frac{\bar{\mu}}{r_2} \quad (6)$$

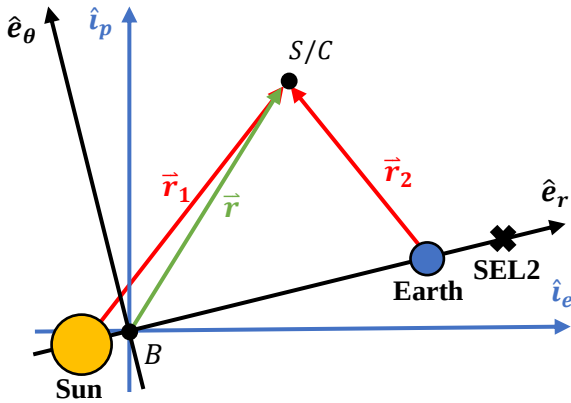


Figure 3. Diagram of the CR3BP System, with the Perifocal frame (black) and Synodic frame (blue).

2.2 Reference Frames

Both MS are initially modelled in the CR3BP domain to access unstable manifolds, however to implement Lambert's problem later, it will be necessary to analyse the motion of the spacecraft in a 2BP domain. The final legs are completed in the heliocentric inertial reference frame, or Ecliptic J2000 frame (abbreviated henceforth as the Sun-Centred Inertial (SCI) frame). A translation of $+\bar{\mu}$ is applied in the x-axis of the spacecraft state space $\mathbf{X}$ to offset the origin of the Synodic frame from the barycentre of the Primaries into a Sun-centred Synodic (SCS) frame:

$${}^{SCS}\mathbf{X} = {}^S\mathbf{X}_x + \bar{\mu} \quad (7)$$

In the case of the EPF, intermediary analysis is also required in that of an Earth-Centred Synodic (ECS) frame, via a translation of $-1 + \bar{\mu}$ in the x-axis of $\mathbf{X}$:

$${}^{ECS}\mathbf{X} = {}^S\mathbf{X}_x - 1 + \bar{\mu} \quad (8)$$

The ephemeris time of rotation and the corresponding State Transformation Matrix (STM) $[IS]$ defining the transformation from a body-centred Synodic frame are gathered to retrieve the spacecraft state in the SCI frame:

$${}^{SCI}\mathbf{X} = {}^{SCS,ECS}\mathbf{X}[IS] \quad (9)$$

2.2.1 Jacobi Integral

The non-dimensional Jacobi integral in the synodic frame is defined as [6]:

$$C = \Omega - \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \quad (10)$$

Where C is the constant of integration (or the Jacobi constant), and $(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$ is the square root magnitude of the velocity term. This is used as a validation of the CR3BP EOM integration in this study.

2.3 Locating Equilibrium Points

Equilibrium points are positions where the gravitational and centrifugal accelerations of the primaries cancel out, allowing the spacecraft to appear stationary relative to the Synodic frame. Since equilibrium points are particular solutions of the CR3BP dynamical system that do not vary with time, they are locations in space where the spacecraft has zero velocity and acceleration relative to the primaries. This allows the following simplifications to be made:

$$\dot{x} = \dot{y} = \dot{z} = \ddot{x} = \ddot{y} = \ddot{z} = 0 \quad (11)$$

All equilibrium points in the CR3BP lie in the x-y plane, such that the 3 collinear equilibrium points must have $y = z = 0$, and can be returned by determining the roots of a 5th order polynomial in the x-plane only. In these cases, iterative methods such as the Newton-Raphson method can be applied [6].

2.4 Escaping Equilibrium Points

The EOM can be linearised about the equilibrium point and eigenvectors of the system computed, which help to shed light on the stability characteristics in the neighbourhood of the equilibrium point. To do so the Jacobian matrix, a matrix of all 1st order partial derivatives is evaluated at SEL2, and the eigenvector of the system returned.

Each of the collinear equilibrium points eigenvalues include both one purely positive real and one purely negative real component, concluding that they are unstable. Practically, this implies that if the spacecraft is placed at rest in the vicinity of one of these points, it will exit the equilibrium point over time. This phenomenon is exploited this study, by using the exponential growth component to perturb the spacecraft slightly from SEL2 to begin its escape trajectory via invariant manifolds.

Invariant manifolds are chaotic dynamical channels that are only available in mission design when analysing 3-body systems or higher and can provide low-cost transport to the outermost reaches of the solar system.

Invariant manifolds are divided into stable or unstable manifolds, dictating trajectory convergence to or divergence from equilibrium points respectively. The unstable manifold is used in all designs in this study accessible by treating SEL2 as the starting point vector $\mathbf{X}^*$ and either adding or subtracting unstable eigenvector $\mathbf{V}_{unstable}$, multiplied by scaling factor ε to create a new Initial Condition (IC) (plotted in Figure 4) [7]:

$$\mathbf{IC}_1 = \mathbf{X}^* + \varepsilon \mathbf{V}_{unstable} \quad (12)$$

$$\mathbf{IC}_2 = \mathbf{X}^* - \varepsilon \mathbf{V}_{unstable} \quad (13)$$

The scaling factor represents the magnitude of the perturbation, which is subject to the 3-body system being analysed. The unstable manifold is used in all designs in this study. In MS 1, a new $\mathbf{IC}_1$ helps the spacecraft escape SEL2 by drifting into deep space along the exterior manifold and enter heliocentric orbit at the Aphelion. In MS 2, a new $\mathbf{IC}_2$ in the opposing direction allows the spacecraft to escape from SEL2 along the interior manifold, converging towards the Perigee.

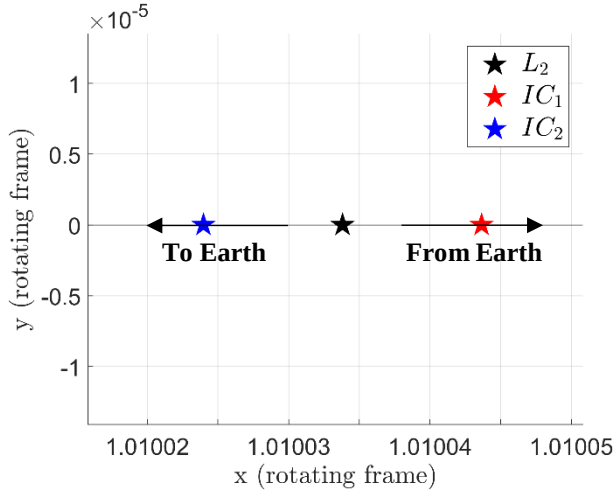


Figure 4. SEL2 and perturbed Initial Conditions.

3. Asteroid Search

NEOs are objects with a Perihelion radius of 1.15 AU or lower [4]. When pruning asteroids, reachability and scientific value are considered. The MOID is the absolute minimum Euclidean distance between a point on a spacecraft orbit and a point on a given asteroid orbit [5] and hence, is a measure of the closest possible approach of the two (depicted in Figure 5). The MOID in this study removes any asteroids that do not come within a given distance to the spacecraft orbit over a 2-year duration. Filters on highly inclined asteroids also removes potentially costly transfers. Once the transfer is deemed achievable, the Absolute Magnitude (H) of the asteroid can be used to weigh up the candidate's scientific value, an indicator of diameter subject to albedo (its reflectiveness).

3.1 MOID Filtering

The MOID was derived by completing an exact geometrical analysis. This approach is achievable using only the orbital elements (OEs) of the respective orbits $\overline{\mathbf{OE}}_{1,2}$:

1. Using $\overline{\mathbf{OE}}_{1,2}$, retrieve the respective position $\mathbf{r}_{1,2}$ and velocity $\mathbf{v}_{1,2}$ vectors.
2. For each state vector, calculate the Angular momentum $\mathbf{h}_{1,2}$ and Eccentricity vectors $\mathbf{e}_{1,2}$:

$$\mathbf{h}_{1,2} = \mathbf{r}_{1,2} \times \mathbf{v}_{1,2} \quad (14)$$

$$\mathbf{e}_{1,2} = \frac{1}{\mu} (\mathbf{v}_{1,2} \times \mathbf{h}_{1,2}) - \frac{\mathbf{r}_{1,2}}{r_{1,2}} \quad (15)$$

3. Define the Perifocal unit vectors $\hat{\mathbf{i}}_{e1,2}$, $\hat{\mathbf{i}}_{p1,2}$, $\hat{\mathbf{i}}_{h1,2}$ for each orbit.
4. Determine the mutual nodal line $\mathbf{K}$ of the two reference frames:

$$\mathbf{K} = \frac{\hat{\mathbf{i}}_{h1} \times \hat{\mathbf{i}}_{h2}}{\|\hat{\mathbf{i}}_{h1} \times \hat{\mathbf{i}}_{h2}\|} \quad (16)$$

5. For each orbit, calculate the true anomalies in each half of the respective orbit:

$$\theta_{1a} = \text{atan2}(\mathbf{K} \cdot \hat{\mathbf{i}}_{p1}, \mathbf{K} \cdot \hat{\mathbf{i}}_{e1}) \quad (17)$$

$$\theta_{1b} = \theta_{1a} + \pi \quad (18)$$

$$\theta_{2a} = \text{atan2}(\mathbf{K} \cdot \hat{\mathbf{i}}_{p2}, \mathbf{K} \cdot \hat{\mathbf{i}}_{e2}) \quad (19)$$

$$\theta_{2b} = \theta_{2a} + \pi \quad (20)$$

6. Determine position vectors $\mathbf{r}_{1a}$, $\mathbf{r}_{1b}$, $\mathbf{r}_{2a}$ and $\mathbf{r}_{2b}$ for each of the true anomalies.
7. Within the set of position vectors, there are now 4 possible distances for which one of them is the MOID. Calculate the Euclidean distance for each possibility:

$$d_1 = \|\mathbf{r}_{1a} - \mathbf{r}_{2a}\| \quad (21)$$

$$d_2 = \|\mathbf{r}_{1a} - \mathbf{r}_{2b}\| \quad (22)$$

$$d_3 = \|\mathbf{r}_{1b} - \mathbf{r}_{2a}\| \quad (23)$$

$$d_4 = \|\mathbf{r}_{1b} - \mathbf{r}_{2b}\| \quad (24)$$

8. Return the minimum solution of the distances.

3.2 Lambert's Problem

The goal of Lambert's Problem (or orbital boundary value problem) is to find the connecting orbit between two points in space in a given time. If given the initial $\mathbf{r}_{S/C}$ and final $\mathbf{r}_{NEO}$ position vectors, the transfer time Δt between these position vectors, solving Lambert's Problem defines the orbital elements of the transfer orbit and therefore both the initial $\mathbf{v}_0$ and final $\mathbf{v}_f$ velocity vectors for the transfer leg may be determined. In the end for this study Izzo's method [9] was used for its low

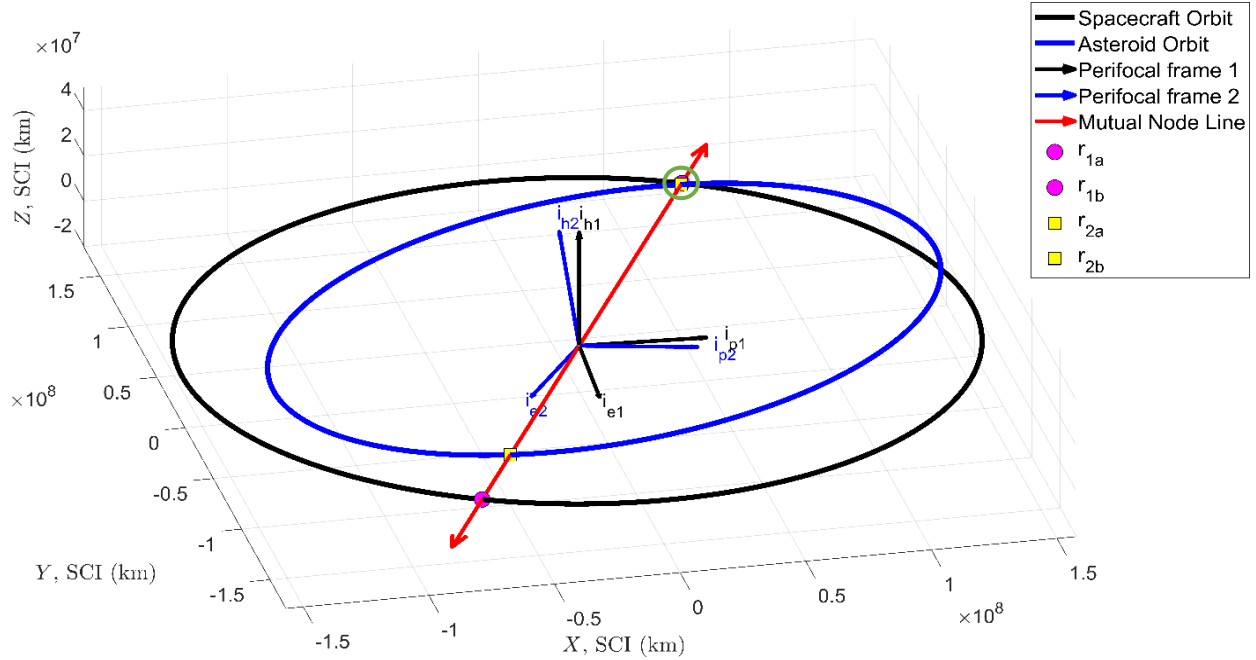


Figure 5. Orbit plot demonstrating the MOID (circled) between the spacecraft and a NEO orbit at a given date.

computational cost; however, in earlier stages the Universal Variables formulation (further explained by Bate et al [10] and a systematic guide provided by Davis [11]) proved equally dependable.

3.3 Pork chop Plots

Pork chop plots are contour plots (their name inspired by their shape) of launch dates versus arrival dates for transfer trajectories between two locations and provides mission designers with a graphical representation of solutions to Lambert's problem to serve as a preliminary orbit determination tool. Pork Chop plots commonly depict useful figures of merit such as departure and arrival v_∞ values, which are derivable from the Lambert solver velocity outputs and velocities of the departing or approaching orbit or body [8]:

$$v_\infty^{dep} = |v_0 - v_{sc}| \quad (25)$$

$$v_\infty^{arr} = |v_{ast} - v_f| \quad (26)$$

Where v_{sc} and v_{ast} are the orbital velocities of the spacecraft and asteroid in the SCI frame respectively. By applying the Lambert solver, it is possible to determine a vast range of potential transfer trajectories for each combination of departure and arrival date.

Porkchop plots were built in tandem with the asteroid search to depict candidates that met the departure and arrival velocity mission constraints. A TOF limit was also set to remove very distant NEOs that were most likely irredeemably out of bounds of the TOF mission requirement, but still allow breathing room for the optimisation scheme later.

4 Earth Powered Flybys

Planetary flybys are often used in mission designs to 'steal' some of a given planet's energy (negligible to the planet but by no means minor to the thief) during a hyperbolic flyby, returning a higher outbound hyperbolic spacecraft velocity relative to the Sun than the inbound trajectory prior. An EPF is the main focal setup for MS 2, utilised both as a means to counteract the longer path taken in MS 1 over the manifold transfer to the Aphelion and gain a free boost from the Earth. For more detail in the hyperbolic EPF formulae outlined in this section, see Battin [12].

The proposal of the EPF is to remove the enforced straight to a given NEO in heliocentric orbit. The core idea of the EPF is broken into 2 legs: a transfer from Earth to NEO, and an escape trajectory from Perigee to Earth Sphere of Influence (SOI). Once leg 1 completes a Lambert analysis to a given asteroid directly from Earth, then under the Patched-Conics approximations it is then possible to forget about the heliocentric NEO transfer and focus solely on the gravitational interaction between the Earth and spacecraft in a 2BP domain of the SOI. Coupling with the knowledge that the spacecraft must leave the Earth SOI with an excess hyperbolic velocity equal to that of the outbound flyby velocity, then the aim of the analysis is to find the hyperbolic trajectory from the Spacecraft at Perigee to the SOI edge that returns the velocity equality within the SCI frame. The outputs of leg 1 are used to calculate $^\oplus v_\infty^+$ and $^\oplus v_\infty^{arr}$ from Equations (25) and (26) respectively, replacing v_{sc} with $v_\oplus$, the orbital velocity of the Earth. $^\oplus v_\infty^+$ is then used to determine the burn required to get to the same asteroid from the spacecraft. Currently the state of the spacecraft

at Perigee is undefined due to an unknown ephemeris time, so this is calculated a posteriori:

1. Calculate semi major axis at hyperbola, a_∞ :

$$a_\infty = -\frac{\mu_\oplus}{|\oplus \mathbf{v}_\infty^+|^2} \quad (27)$$

2. Calculate the hyperbolic eccentricity, e_∞ :

$$e_\infty = 1 - \frac{r_{\text{Perigee}}}{a_\infty} \quad (28)$$

3. Calculate the true anomaly of the spacecraft θ_∞ , when it passes the Earth SOI boundary:

$$\theta_\infty = \cos^{-1} \left(\frac{a_\infty(1 - e_\infty) - r_\infty^{\text{SOI}}}{e_\infty r_\infty^{\text{SOI}}} \right) \quad (29)$$

Where r_∞^{SOI} is the radius of the Earth SOI.

4. Calculate the hyperbolic anomaly of the spacecraft H_∞ :

$$H_\infty = \tanh^{-1} \left(\sqrt{\frac{e_\infty - 1}{e_\infty + 1}} \tanh \left(\frac{\theta_\infty}{2} \right) \right) \quad (30)$$

5. Calculate the TOF of the spacecraft to travel from the Perigee to the Earth SOI Edge, T_∞ :

$$T_\infty = \sqrt{\frac{(-a_\infty)^3}{\mu_\oplus}} (e_\infty \sinh(H_\infty) - H_\infty) \quad (31)$$

6. Calculate the true ephemeris time at which the spacecraft is at Perigee, et_{Perigee} :

$$et_{\text{Perigee}} = et_{\text{dep}} - T_\infty \quad (32)$$

Where et_{dep} is the ephemeris time of departure used in Leg 1.

7. Rotate the spacecraft state at the Perigee from the Synodic frame to the SCI frame using et_{Perigee} .

8. Define the Perifocal frame of the hyperbolic trajectory using $\mathbf{r}_{\text{SCI}}^{\text{Perigee}}$:

$$\hat{\mathbf{e}} = \frac{\mathbf{r}_{\text{SCI}}^{\text{Perigee}}}{\|\mathbf{r}_{\text{SCI}}^{\text{Perigee}}\|} \quad (33)$$

$$\hat{\mathbf{h}} = \frac{\mathbf{r}_{\text{SCI}}^{\text{Perigee}} \times \oplus \mathbf{v}_\infty^+}{\|\mathbf{r}_{\text{SCI}}^{\text{Perigee}} \times \oplus \mathbf{v}_\infty^+\|} \quad (34)$$

$$\hat{\mathbf{p}} = \hat{\mathbf{e}} \times \hat{\mathbf{h}} \quad (35)$$

9. Calculate the velocity of the spacecraft on hyperbolic trajectory at Perigee, V_{Perigee} :

$$V_{\text{Perigee}} = \sqrt{2 \left(-\frac{\mu_\oplus}{2a_\infty} + \frac{\mu_\oplus}{\|\mathbf{r}_{\text{SCI}}^{\text{Perigee}}\|} \right)} \quad (36)$$

10. Calculate the inertial velocity vector of the spacecraft on hyperbolic trajectory at Perigee in the SCI frame, ${}^{\text{SCI}}\mathbf{v}_\infty^+$:

$${}^{\text{SCI}}\mathbf{v}_\infty^+ = V_{\text{Perigee}} \hat{\mathbf{p}} \quad (37)$$

11. Return the delta-v required for the spacecraft to insert into the hyperbolic EPF trajectory and transfer directly to the given NEA, $\Delta \mathbf{V}_1$:

$$\Delta \mathbf{V}_1 = {}^{\text{SCI}}\mathbf{v}_\infty^+ - \mathbf{v}_{\text{SCI}}^{\text{Perigee}} \quad (38)$$

Now effectively $\Delta \mathbf{V}_1 = \mathbf{v}_\infty^{\text{dep}}$. Similarly to Design 1, this approach can be cast over a range of departure and arrival dates to return Pork chop plots of potential candidates.

5 Trajectory Optimisation

In preliminary mission design, trajectory optimisation is typically saved for last, as the initial guess must be sufficiently accurate to return a scenario-specific minimum. This initial guess is the culmination of all the previous topics, and includes parameters required for constraints and the cost functions in question. The cost functions are the aforementioned mission requirements. Constraints can be either linear or nonlinear. Linear constraints ensure that the start point of the mission stays as the perturbed SEL2 point vector. Table 1 shows the initial guess for each MS:

Table 1. Trajectory optimisation initial guess vectors

MS 1	MS 2
$\mathbf{IC}_1$	$\mathbf{IC}_2$
TOF_1	TOF_1
$\mathbf{v}_\infty^{\text{dep}}$	$\mathbf{v}_\infty^{\text{dep}}$
TOF_2	TOF_2
$\mathbf{v}_\infty^{\text{arr}}$	TOF_3
JD_{dep}	$\mathbf{v}_\infty^{\text{arr}}$
	JD_{dep}

JD_{dep} allows the optimisation to fluctuate within a day margin. The true beauty behind trajectory optimisation is that constraints previously enforced on the preliminary mission design when generating the initial guess can be relaxed during optimisation. This can have a positive effect whereby the trajectory optimisation scheme finds a better trajectory because of this. For example, the optimiser may be able to find superior trajectories that may not encounter the Aphelion or the Perigee, and at a reduced cost function.

5.1 Non-Linear Programming via Fmincon

Fmincon stands for constrained minimisation of a function, a Non-Linear Programming (NLP) solver available through MATLAB. This solver also accommodates some common algorithms including the interior-point [13] and Sequential Quadratic Programming (SQP) [14] techniques, which are both applied throughout optimisation in this study to a single shooting method.

The single shooting transcription process works by modelling the trajectory as one continuous function, which includes both the manifold insertion in the CR3BP domain, and asteroid transfer in the SCI domain for each design. The total trajectory is iteratively simulated by integrating the corresponding EOM, for which either the

velocity or the TOF cost functions are minimised throughout whilst satisfying the constraints and boundary conditions.

6 Mission Scenario 1

The following work presented in this chapter was completed in MATLAB. Other tools included the JPL Horizons SBDB [4] for access to all documented asteroids, and NASA's Spice toolkit [15] for additional syntax.

6.1 CR3BP

A validation of the CR3BP method was completed by computing the Jacobi integral for the length of the manifold transfer from SEL2, calculating Equation (10) for position and velocity vectors at each integration step of the CR3BP EOM. The Jacobi integral over the course of the manifold transfer for MS 1 was $C_1 = 1.5004$.

6.1.1 Sun-Earth Collinear Equilibrium Points

Newton-Raphson method was applied with a tolerance of 1×10^{-10} to locate SEL2 from a given initial guess. The generalised initial guess $\mathbf{X}_{L2}^0$ was as shown in Equation (39) and after 4 iterations returned the SEL2 state point vector $\mathbf{X}_{L2}^*$ in Equation (39):

$$\mathbf{X}_{L2}^0 = [1.02(1 - \bar{\mu}) \quad 0 \quad 0 \quad 0 \quad 0 \quad 0] \quad (39)$$

$$\mathbf{X}_{L2}^* = [1.0100 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0] \quad (40)$$

Generalised initial guesses were repeat for the remaining collinear equilibrium points to ensure consistency.

6.1.2 Perturbation into the Unstable Manifold

The new initial condition was calculated using Equation (12), using the unitised $\mathbf{V}_{unstable}$ eigenvector evaluated at the SEL2 vector $\mathbf{X}_{L2}^*$ and scaling factor of $\varepsilon = 3 \times 10^{-5}$ to return the Initial Condition:

$$\mathbf{IC}_1 = \begin{bmatrix} 1.0010 \\ -5.3624 \times 10^{-6} \\ 0 \\ 2.4434 \times 10^{-5} \\ -1.3323 \times 10^{-5} \\ 0 \end{bmatrix} \quad (41)$$

6.2 Trajectory Propagation

The CR3BP EOM were initialised at Equation (41) and integrated via ode45. An event function was employed to ensure propagation terminated when the spacecraft reached the Aphelion. The event function leaned on the knowledge that the dot product between the relative position vector of the spacecraft with respect to a given primary and inertial velocity vector become zero at the Apoapsis and Periapsis due to vector orthogonality:

$$\vec{r}_{wrt} \cdot \mathbf{v} = 0 \quad (42)$$

Where the relative position vector of the spacecraft with respect to the Sun was:

$$\vec{r}_{wrt} = [x + \bar{\mu} \quad y \quad z] \quad (43)$$

The state space vector at the Aphelion was:

$$\mathbf{X}_{Aphelion} = \begin{bmatrix} 1.0516 \\ -0.2643 \\ 0 \\ -0.0339 \\ -0.1349 \\ 0 \end{bmatrix} \quad (44)$$

Trajectory propagation to the Aphelion terminated at $t_e = 5.9057$. Reintroducing units via TU from Equation (3) and dividing by the length of a sidereal day returns the time taken to complete the escape via manifold insertion, $TOF_1 \cong 343 \text{ days}$, or $\sim 11.43 \text{ months}$. Trajectory propagation was validated by holding the event function and extending the time of integration to return a full L_2 homoclinic orbit.

6.3 Rotating Frames

In preparation for the asteroid search, $\mathbf{X}_{Aphelion}$ was moved to the SCI frame and dimensions returned to km and kms^{-1} for a 2-year range spanning 01/01/26 – 01/01/28, and the spacecraft COE returned for each day step. The SC COE in radians on 01/01/26 was:

$$sciCOE_{S/c} = \begin{bmatrix} 1.6986 \times 10^8 \\ 0.0405 \\ 1.6191 \times 10^{-4} \\ -2.0313 \\ -2.7491 \\ 6.2832 \end{bmatrix} \quad (45)$$

This also provides a simple sanity check since Semi-major axis is in the region of 1 AU.

6.4 Asteroid Searching from Aphelion

The JPL Horizons SBDB data query allows a user to export parameters such as the Absolute Magnitude, Diameter (largely unknowns), and COE for all tracked NEOs to date. The MOID between the spacecraft over the 2 years in day steps and each given NEO of $i \leq 10^\circ$ and $H \leq 25$ was calculated using the process laid out in Section 3.1. Individual NEO state spaces were gathered via https requests directly from NASA's JPL Horizons database using the asteroid's SPK-ID and cspice_furnsh.

6.4.1 MOID Threshold

To return only asteroids reachable within the mission constraints, a MOID threshold was defined based upon the delta-v constraint of $150ms^{-1}$. By adding the maximum allowable delta-v to the velocity state of the spacecraft in Equation (44), a new orbit defining the furthest distance the spacecraft can reach is found. It is simple enough to take the difference between the semi-major axes of the original and new orbits using the Vis-Viva equation, returning a cut-off value of $MOID_{th} = 1.9338 \times 10^6$ (0.0129 AU) for MS 1.

6.4.2 Returning Candidates

The MOID approach in Section 3 was run for all NEO COEs and all spacecraft COEs in the 2-year departure range. In cases that $MOID > MOID_{threshold}$ for a given date, the asteroid was kept within the search, as there may be better dates where it travelled within range. If the $MOID \leq MOID_{threshold}$ at a given date, then the Lambert solver analysis described in Section 6.4.3 was run. The minimum v_{∞}^{dep} transfer trajectory was evaluated against the velocity cut-offs $v_{\infty}^{dep} \leq 160 \text{ m s}^{-1}$ and $v_{\infty}^{arr} \leq 6 \text{ km s}^{-1}$. The ceiling of the mission constraints was raised slightly to allow the optimisation scheme to try retrieving an optimal transfer back within the mission requirements. If the minimum transfer trajectory failed to meet these velocities bounds, the asteroid was removed from the overall search. Otherwise, if the asteroid proved to be a potential candidate, trajectory details and its pork chop plot was stored.

Asteroid searches from the Aphelion in MS 1 returned 71 potential candidates, 31 of which already met the overall v_{∞}^{dep} and v_{∞}^{arr} mission constraints. The largest possible viable candidate was 2004 FM17 with a $H = 19.36$. The shortest transfer was to 2013 XV8 with a $TOF = 32 \text{ days}$. The top 10 candidates shown in Table 2 (5 lowest Absolute Magnitudes and hence potentially largest and 5 shortest transfers) were carried through trajectory optimisation.

Table 2. 10 candidates selected for optimisation in MS1.

SPK-ID	Name	Abs. Mag.	TOF (Days)
3175334	2004 DK1	21.30	220
20387816	2004 FM17	19.36	162
3293282	2005 TF45	23.10	41
3414400	2008 MQ1	24.20	104
3446391	2009 CO5	21.85	454
3654379	2013 XV8	21.89	32
20520585	2014 OA2*	21.48	263
3674937	2014 OV3	23.20	82
3711785	2015 DP155	21.62	300
54426746	2024 CN5	24.98	91

*520585 Saci

6.4.3 Building Pork Chop Plots

Pork chop plots used a 2-year departure window of 01/01/2026 – 01/01/2028 and an arrival window spanning 01/01/2026 – 15/05/2029. For all day steps in departure and arrival positions of the spacecraft and each asteroid in the SCI frame, the Lambert solver was applied to return the initial and final transfer velocity vectors $\mathbf{v}_0$ and $\mathbf{v}_f$ for all possible ballistic transfers of $TOF < 500$ from the Aphelion to the asteroid. Equations (25) and (26) returned v_{∞}^{dep} and v_{∞}^{arr} matrices to be plotted. During this loop, the days of departure and the TOF

matrices were also stored to be used as X-Y planes for the contour plots. Porkchop plots were built to display contours ranging from $0 \text{ km s}^{-1} \leq v_{\infty}^{dep} \leq 1 \text{ km s}^{-1}$ and $0 \text{ km s}^{-1} \leq v_{\infty}^{arr} \leq 10 \text{ km s}^{-1}$, as illustrated in Figure 6 for 2004 FM17.

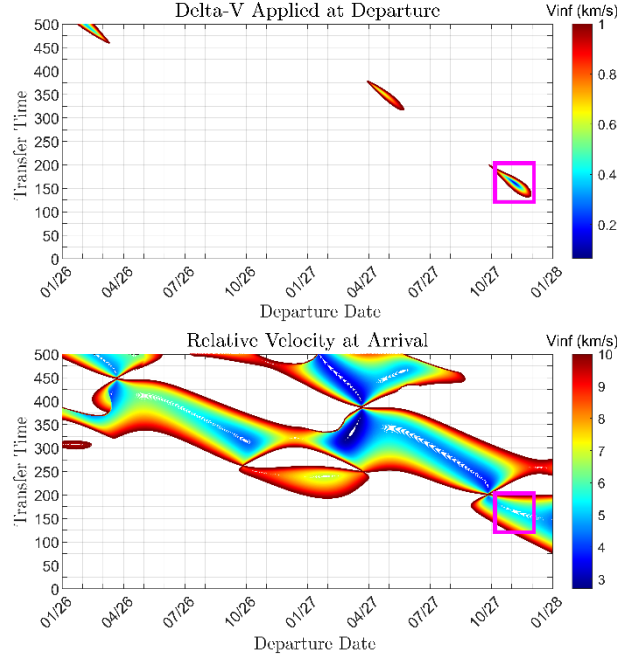


Figure 6. Departure and arrival velocity Porkchop plot for 2004 FM17, and minimum departure delta-v trajectory (centre region of squares).

6.6 Trajectory Optimisation

The single shooting method used a basic fmincon strategy. Simulations were completed in a similar fashion to those described in Section 5. For this section, the notation is altered slightly from Table 1. Figure 7 shows a cartoon of the simulation carried out to assist the reader.

$$\mathbf{X} = \begin{bmatrix} \mathbf{IC}_1 \\ TOF_1 \\ \mathbf{v}_{\infty}^{dep} \\ TOF_2 \\ \mathbf{v}_{\infty}^{arr} \\ JD_{dep} \end{bmatrix} \rightarrow \mathbf{X} = \begin{bmatrix} \mathbf{X}_0 \\ TOF_1 \\ \Delta \mathbf{V}_1 \\ TOF_2 \\ \Delta \mathbf{V}_2 \\ JD_{dep} \end{bmatrix} \quad (46)$$

$\mathbf{X}_0$ was integrated using the CR3BP EOM for the length of TOF_1 . The last step in integration was rotated into the SCI frame to return the state space vector at the end of the escape from SEL2, $\mathbf{X}_{Esc}$. $\Delta \mathbf{V}_1$ was added to the velocity component of this state space vector to create the state space vector at departure $\mathbf{X}_{Dep}$:

$$\mathbf{X}_{Dep} = \mathbf{X}_{Esc} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \Delta \mathbf{V}_1 \end{bmatrix} \quad (47)$$

$\mathbf{X}_{Dep}$ was then integrated using the 2BP EOM for the length of TOF_2 , and when solver reached a local minimum last step in integration returned the state space vector $\mathbf{X}_{Arr}$ for asteroid rendezvous (otherwise undershooting, overshooting, or not meeting the constraints).

The cost function (shown in Table 3), algorithm (interior-point or SQP), and constraints were all selected, and fmincon was run iteratively until convergence

Table 3. Cost functions (J) used in MS 1.

J	$F(J)$
Total delta-v	$\ \Delta\mathbf{V}_1\ + \ \Delta\mathbf{V}_2\ $
Total TOF	$TOF_1 + TOF_2$
Departure delta-v	$\ \Delta\mathbf{V}_1\ $
Arrival delta-v	$\ \Delta\mathbf{V}_2\ $

This process was looped, such that optimisation could be optionally rerun with different settings and with the output of the prior optimisation as the next initial guess. This allowed the flexibility to add constraints in to be added in over iterations or alter the course of action if the solver failed to find a minimum. Output functions were applied to generate trajectories for final candidates. The data for optimisation was normalised throughout to assist the solver tackle large differences between the initial guess values.

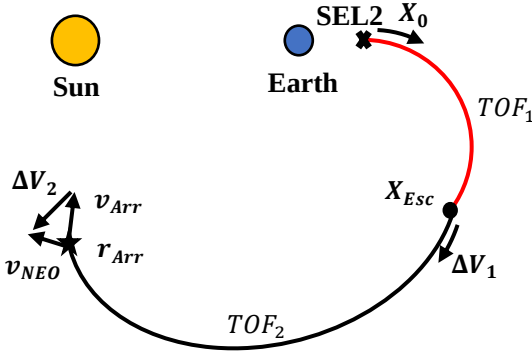


Figure 7. Single shooting method for MS 1.

6.6.1 Boundary Conditions

Lower **lb** boundary conditions for TOF values were set to be positive $\mathbf{lb}(TOF_1, TOF_2) = 0$ to mitigate infeasible transfers. J_{Dep} was set within ± 1 day of departure (dep), such that $\mathbf{lb}(J_{Dep}) = J_{Dep} - dep$ and $\mathbf{lb}(J_{Dep}) = J_{Dep} + dep$. This helpful limitation is possible because the Pork Chop plot building process assesses all possible transfers and returns the minimum $\mathbf{v}_{\infty}^{dep}$ trajectory within a *daily* resolution. Therefore, it is expected that in cases of trajectories retrieved pre-optimisation of $\mathbf{v}_{\infty}^{dep} < 150\text{ms}^{-1}$, a trajectory of $\mathbf{v}_{\infty}^{dep} = 150\text{ms}^{-1}$ is discoverable through sufficient optimisation

within a 1 day range of search. With the aid of constraints, surrounding mission constraints such as $\mathbf{v}_{\infty}^{arr}$ and TOF can then also be minimised by maxing out the remaining available delta-v. In cases the initial guess has $\mathbf{v}_{\infty}^{dep} > 150\text{ms}^{-1}$, the optimiser will search for transfers that reduce this at the cost of the other mission requirements.

Otherwise, both **lb** and **ub** conditions were set to $-\infty$ and $+\infty$ for all remaining components of the initial guess to allow the solver to explore freely.

6.6.2 Constraints

A linear equality constraint ensured that the mission begun from SEL2 of form $\mathbf{A}_{eq}\mathbf{X} = \mathbf{b}_{eq}$. Nonlinear equality $\mathbf{c}_{eq}(\mathbf{x})$ and inequality $\mathbf{c}(\mathbf{x})$ constraints are functions that return vectors. The fundamental nonlinear equality constraint was:

$$\mathbf{c}_{eq}(1) = \mathbf{r}_{Arr}(TOF_2) - \mathbf{r}_{NEO}(TOF_2) \quad (48)$$

Such that the difference between the position vector of the spacecraft and the position vector of the asteroid at the end of the transfer equalled zero. A second mandatory constraint fixed relative velocity alignment at arrival:

$$\mathbf{c}_{eq}(2) = \mathbf{v}_{Arr}(TOF_2) + \Delta\mathbf{V}_2 - \mathbf{v}_{NEO}(TOF_2) \quad (49)$$

These three constraints defined the borders for the start and end state space of the mission.

Mentioned back in Section 5, an optional constraint to add (or thereby remove) forced entry to the Aphelion was added via reapplication of Equations (42) and (43):

$$\mathbf{c}_{eq}(3) = \vec{\mathbf{r}}_{wrt}(TOF_1) \cdot \mathbf{v}(TOF_1) \quad (50)$$

Last were 2 nonlinear inequality constraints for departure and approach velocity requirements:

$$\mathbf{c}(1) = \|\Delta\mathbf{V}_1\| - \frac{0.15}{LU} TU \quad (51)$$

$$\mathbf{c}(2) = \|\Delta\mathbf{V}_2\| - \frac{6.0}{LU} TU \quad (52)$$

6.7 Final Candidates

The optimisation scheme was run for all asteroids in Table 2, with the Aphelion entry removed in all cases. Generally, the best set of iterations found was to first minimise $\Delta\mathbf{V}_2$ whilst adding in the mandatory constraints, $\mathbf{c}(1)$ in Equation (51), and alleviating $\mathbf{c}_{eq}(3)$ in Equation (50). Once a local minima was returned, the cost function was switched to total TOF, and all constraints were set aside from $\mathbf{c}_{eq}(3)$ once again. Thus, the 2nd loop would begin with at least 2 mission requirements met already (or at least with $\Delta\mathbf{V}_2$ as low as possible) and attempt to reel in the TOF as much as possible by pushing up to the ceilings of velocities $\Delta\mathbf{V}_1$ and $\Delta\mathbf{V}_2$. The interior-point algorithm proved to be the most effective for MS 1. Table 4 shows the results of trajectory optimisations (see Figure 8 for an optimised scenario) and the mission requirement difference prior to optimisation.

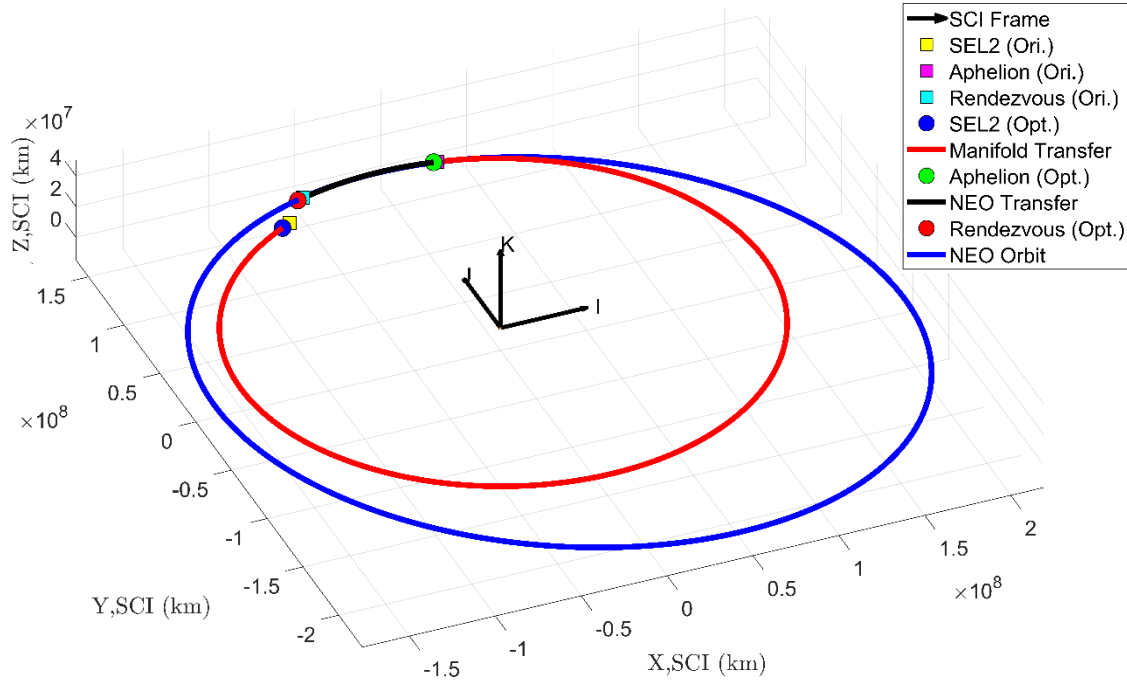


Figure 8. Complete MS 1 trajectory from SEL2 to 2013 XV8 via a singular burn at the Aphelion, in the SCI frame.

Table 4. Trajectory optimisation results for MS 1.

Name	TOF_{tot} (Days)	TOF_{tot} (Mo.)	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})	Name	TOF_{tot} (Days)	TOF_{tot} (Mo.)	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})
2004 FM17	502.13 [-3.18]*	16.74 [-0.11]	0.1500 [+0.0896]	5.3282 [+0.0493]	2013 XV8	376.22 [+0.91]	12.54 [+0.01]	0.1500 [+0.0116]	3.3616 [-0.1340]
2004 DK1	558.78 [-4.53]	18.63 [-0.15]	0.1500 [+0.0993]	5.4206 [-0.1025]	2005 TF45	382.25 [-2.06]	12.74 [-0.07]	0.1500 [+0.0457]	3.8592 [+0.1248]
2014 OA2	606.60 [+0.29]	20.22 [+0.01]	0.1500 [+0.1122]	4.9677 [-0.0515]	2014 OV3	422.63 [-2.68]	14.09 [-0.09]	0.1500 [+0.0228]	4.5634 [+0.1229]
2015 DP155	639.28 [-4.03]	21.31 [-0.13]	0.1499 [+0.1384]	4.2687 [-0.0973]	2024 CN5	429.48 [-4.83]	14.32 [-0.16]	0.1500 [+0.1263]	4.247 [+0.1052]
2009 CO5	797.59 [+0.28]	26.59 [+0.01]	0.1499 [+0.0289]	4.9666 [-0.0175]	2008 MQ1	451.25 [+3.94]	15.04 [+0.13]	0.1499 [+0.0105]	4.3766 [-0.0545]

*Increase [+] or decrease [-] from original trajectories. Un-highlighted results were optimised for TOF. Highlighted results were optimised for relative velocity at arrival.

7 Mission Scenario 2

7.1 CR3BP

A similar Jacobi integral validation was run for MS 2, returning an expected value of $C_2 = C_1 = 1.5004$, due to a duplicate initialisation, just in the opposed direction.

7.1.1 Returning to Earth

IC_2 was generated using Equation (13) perturbed in the opposite direction to IC_1 :

$$IC_2 = \begin{bmatrix} 1.0100 \\ -5.3624 \times 10^{-6} \\ 0 \\ 2.4434 \times 10^{-5} \\ -1.3323 \times 10^{-5} \\ 0 \end{bmatrix} \quad (53)$$

The CR3BP EOM were propagated with an event function to terminate integration at the Perigee, exploiting Equation (42) once more, in this case with the position vector of the spacecraft with respect to the Earth:

$$\vec{r}_{wrt} = [x - (1 - \bar{\mu}) \quad y \quad z] \quad (54)$$

The state space vector at the Perigee was:

$$\mathbf{X}_{\text{Perigee}} = \begin{bmatrix} 0.9908 \\ 3.6920 \times 10^{-4} \\ 0 \\ -0.0202 \\ -0.0614 \\ 0 \end{bmatrix} \quad (55)$$

The time of termination for escape from SEL2 to the Perigee was $t_e = 2.6218$, or after reintroducing dimensions, $TOF_1 \cong 152 \text{ days, or } \sim 5.07 \text{ months}$. Note that by opting for a manifold transfer to the interior realm, TOF_1 for MS 2 is less than half that of MS 1.

7.2 Asteroid Searching from Perigee

For MS 2, MOID approach was completed for all NEO COEs and all Earth COEs over 2 years in day resolution. In consideration of the potential benefits of the EPF, the threshold was increased to $MOID_{th} = 19 \times 10^6 (0.12701 \text{ AU})$, and departure velocity cutoff to $v_{\infty}^{dep} \leq 300 \text{ ms}^{-1}$. v_{∞}^{arr} and TOF thresholds were kept the same as MS 1. When $MOID \leq MOID_{threshold}$ for a given NEO and date, the process described in Section 4 was carried out. The Lambert solver analysis now assessed all possible transfers using Earth positions instead of spacecraft positions, up to the same TOF cut-off and over the same departure and arrival windows as MS 1. For each transfer, $\oplus v_{\infty}^{+}$ and $\oplus v_{\infty}^{arr}$ are calculated from the Lambert outputs, and $\oplus v_{\infty}^{+}$ is used with Equations (27) – (38) to return v_{∞}^{dep} - and $\oplus v_{\infty}^{arr}$ is treated as $\oplus v_{\infty}^{arr} \cong v_{\infty}^{arr}$. From here, the same top-level process as MS 1 is applied.

7.2.1 Preliminary Candidates

20 NEOs were returned from the asteroid pruning process at Perigee in MS 2, shown in Table 5. None initially met the overall v_{∞}^{dep} limits; prior to optimisation, 9 met the v_{∞}^{arr} mission constraints and 1 was reachable within the TOF requirements of 9 months. With a smaller sample size, all 20 were routed through to optimisation. 2016 HO3 (see Figure 9) is a notable candidate; a small asteroid ($< 60 \text{ m}$) in super-fast rotation state ($\sim 28 \text{ min}$), had already been the topic of concept studies such as NEACO [16] and target of China's Tianwen-2 asteroid sample-return mission in May 2025 [17].

Table 5. 20 preliminary candidates found through MS 2.

SPK-ID	Name	Abs. Mag.	TOF (Days)
3054374	2000 SG344	24.70	445
3153530	2003 HT42	24.80	248
3387534	2007 TH3	24.70	274
3389488	2007 UP6	23.00	203
3394778	2007 WA	24.70	223
3457386	2009 HC	24.70	221
3508128	2010 CO44	24.60	435

3599482	2012 DK4	23.60	268
3710229	2015 CW13	22.90	433
3727176	2015 RF36	23.40	289
3752012	2016 GW221	24.76	305
3770151	2017 DB38	24.00	226
3789720	2017 WJ16	24.45	287
3827152	2018 RC	24.93	313
54192165	2021 RS	24.94	224
54298612	2022 QM6	23.74	112
54417170	2023 XM15	24.21	303
54423681	2024 BV5	24.21	268
20469219	2016 HO3*	24.33	138
20510529	2012 EY11	21.58	234

*469219 Kamo'oalewa

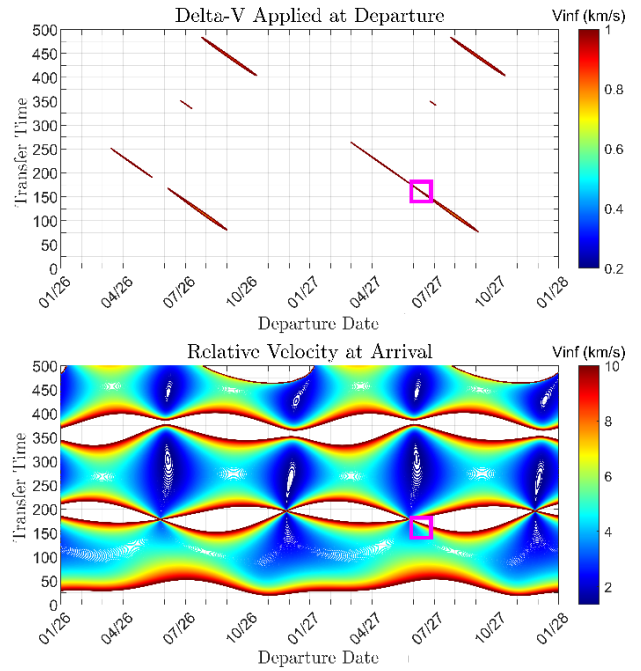


Figure 9. Departure and arrival velocity Porkchop plot for 469219 Kamo'oalewa (2016 HO3), and minimum departure delta-v trajectory (centre region of squares).

7.3 Trajectory Optimisation

The same methodology for MS 1 was followed; again, the notation is altered slightly from Table 1:

$$\mathbf{X} = \begin{bmatrix} \mathbf{IC}_2 \\ TOF_1 \\ v_{\infty}^{dep} \\ TOF_2 \\ TOF_3 \\ v_{\infty}^{arr} \\ JD_{dep} \end{bmatrix} \rightarrow \mathbf{X} = \begin{bmatrix} \mathbf{IC}_2 \\ TOF_1 \\ \Delta \mathbf{V}_1 \\ TOF_2 \\ TOF_3 \\ \Delta \mathbf{V}_3 \\ JD_{dep} \end{bmatrix} \quad (56)$$

In all, the initial guess was a 16x1 vector. Since an additional leg was added, the cost functions considered changed to Table 6:

Table 6. Cost functions (J) used in MS 2.

J	$F(J)$
Total delta-v	$\ \Delta V_1\ + \ \Delta V_2\ $
Total TOF	$TOF_1 + TOF_2 + TOF_3$
Departure delta-v	$\ \Delta V_1\ $
Arrival delta-v	$\ \Delta V_2\ $

Leg 1 was completed in the same manner as that in MS 1. To simulate Leg 2 sufficiently, the gravitational influence of the Earth whilst still in the SOI had to be accounted for. Hence, the intermediary connecting leg was added to ensure that the SOI boundary had been either reached or exceeded in the CR3BP domain, before continuing in the 2BP domain (see Figure 10). The delta-v calculated in Equation (38) was rotated back to the Synodic frame and added to the state space at Perigee and propagated until at or outside the Earth's SOI using the CR3BP EOM once more. Now that the Earth's influence could be considered dismissible, X_{SOI} was rotated back to the SCI frame before integration in the 2BP domain up until rendezvous with the NEO.

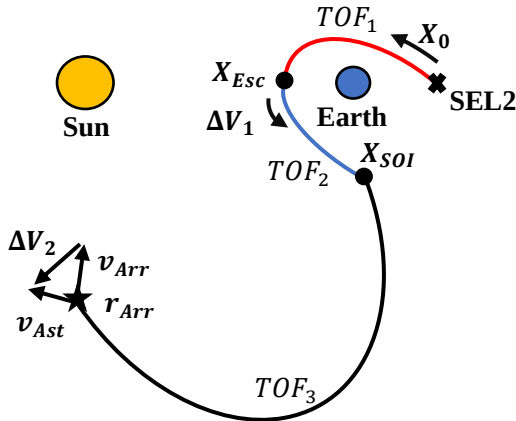


Figure 10. Single shooting method for MS 2.

7.3.1 Boundary Conditions

Boundary conditions were all kept the same, simply accounting for the new vector length as well as setting $lb(TOF_3) = 0$.

7.3.2 Constraints

The linear equality constraint for SEL2 was maintained. Fundamental constraints were all also kept and updated to reflect the new initial guess parameters. The optional constraint was now in relation to the Perigee, by replacing $\vec{r}_{wrt}$ in Equation (50) with Equation (54). A new linear inequality constraint was defined to ensure that the Earth's SOI r_{SOI} was left by the end of Leg 2:

$$c(3) = r_{SOI} - \|r_{s/c}(TOF_2)\| \quad (57)$$

7.4 Final Candidates

Of the 20 preliminary NEOs pulled through to optimisation for MS 2, 50% converged to alternative solutions. Of these 10, 6 returned departure delta-v values greater than $500ms^{-1}$, and thus dismissed. Table 7 shows the remaining 4 candidates that converged to potentially promising alternative trajectories.

Trajectory optimisation was solely spent minimising departure delta-v $\|\Delta V_1\|$, as the majority of original minimum delta-v transfer trajectories fell in the upper bounds of the $300ms^{-1}$ threshold for MS 2. Candidate 2021 RS was the only one to fall within both desired delta-v and relative arrival velocity constraints post optimisation, whilst the others instead had an adverse effect to the expected output, diverging from local minima.

Table 7. Trajectory optimisation results for MS 2.

Name	TOF_{tot} (Days)	TOF_{tot} (Mo.)	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})
2016 HO3	321.71 [+31.39]*	10.72 [+1.05]	0.2567 [+0.0126]	5.1457 [-0.0568]
2022 QM6	295.51 [+30.77]	9.85 [+1.03]	0.4604 [+0.1855]	5.7791 [+0.3924]
2021 RS	413.1047 [+16.52]	13.77 [+0.55]	0.1300 [-0.1311]	2.441 [+0.1346]
2015 RF36	457.57 [+15.72]	15.25 [+0.52]	0.2083 [-0.0694]	4.2703 [+0.4227]

*Increase [+] or decrease [-] from original trajectories

8. Discussion

The initial escape from SEL2 was first implemented with the departure velocity solely in mind, since this was presumed to be the most challenging requirement to meet. This supposition has failed to remain intact when looking at the total final candidates across both mission scenarios. Overall, it has been revealed that both the velocity constraints are relatively attainable, whereas a TOF of sub-9 months has been by far the most elusive.

Naturally, the manifold transfer was considered the most sensible choice when saving fuel – if the spacecraft had the correct equipment (e.g., reaction wheels), it could indeed escape from SEL2 via a slight perturbation and save all remaining leftover delta-v for a singular burn to a NEO once in a more preferable location such as the Aphelion. In the final leg in the designs to rendezvous, the duration of this transfer is not ultimately controllable as it is largely dependent on the asteroids orbital motion. The shortcoming of MS 1 is therefore the TOF of ~ 343 days remained in the slow drift of the unstable manifold up to the Aphelion. The only way to mitigate this would be to depart from this trajectory sooner, which optimisation demonstrated that when the forced Aphelion

entry was removed, could only be done up to a maximum of several days across the batch of 10.

Generally, MS 1 is more than capable of meeting both the delta-v and the relative arrival velocity requirements, but already exceeds a 9 months TOF by the end of the 1st leg. Optimisation results also implied that its simplicity offered a certain rigidity in its design, yet it could be possible that the initial guesses were relatively close to the optimal trajectory within the constraints set, due to its deterministic nature. Despite these critiques, the number of preliminary candidates returned from MS 1 was plentiful considering the restrictions involved in the mission scenarios. It can be determined that whilst MS 1 fell short in its time to reach optimal locations to initiate a final transfer, it had a better NEO reachability overall.

MS 2 was implemented as an alternative to MS 1 for numerous reasons. Primarily it acted to combat the lengthy 1st leg of MS 1, whilst still exploiting a manifold transfer. Secondly from an industrial insight, it was discovered that communications equipment for spacecraft stationed at SEL2 might not be designed to extend out further into deep space, potentially creating blind spots in GNC. Lastly, this scenario gave way to utilise a powered flyby. The resultant escaping leg proved MS 2 to be worthwhile investment, returning a manifold transfer less than half that of MS 1 with a duration of ~152 days, leaving ~4 months for transfers to NEOs within the missions TOF requirements.

Currently, the overruling shortfall of MS 2 is the optimisation, requiring further investigation. The methodology in building the initial guess is sound – leg 1 proved an effective improvement upon MS 1, and the asteroid search at the Perigee is a creative yet apt approach; assessing reachability of NEOs via patched conics whilst still positioned within the Earth's SOI, a method opted for over a brute force EPF approach to reduce computational costs. Despite this, the majority of the outputs increased from the initial guess. This is an indication the initial guess was not a feasible route in the first place, and as such the nearest possible route was determined, albeit at slightly elevated parameters. One improvement to the optimisation would be a correction burn at the SOI boundary, for adjustments to outbound

leg post EPF to reduce any sensitivity in the optimisation scheme that may be adding undue propagation error.

The most favoured candidate from Table 8 would likely have to be 2013 XV8, in terms of overall criteria. Not only does it meet both the velocity requirements, but despite the downfalls of the 1st leg in MS 1, the final transfer to the object is remarkably just over a month long. Its Absolute Magnitude is in the lower end of the filters set, placing it in the 110m – 300m probable diameter range subject to albedo [18]. The main contender would have to be include 2004 FM17 as the potentially largest object, ranging anywhere from 330m - 940m in diameter, and reachable within the delta-v limit. If an extended assessment into the optimisation for MS 2 returned more favourable trajectories, then the following could also be placed under consideration: 2022 QM6 for its speedy mission length, 2021 RS for its exceptionally low relative arrival velocity (and therefore rendezvous flyby window), and 2016 HO3, primarily for its contextual relevance and also backed by a short mission length.

MS 1 presents itself as a good baseline to compare against for any future potential scenarios. Mainly this would include a Lunar Gravity Assist, both as a sole manoeuvre and in combination with an EPF. The benefits of this would be to have more control over the radius of the Perigee at which the EPF is performed, gaining increased energy returns on the outbound leg, potentially improving NEO reachability overall. Another natural step to improve the fidelity of this preliminary concept study would be to drill down to the scene of the asteroid flyby window itself, focusing on parameters such as solar illumination and closest approaches.

9. Conclusions

This paper has provided a concept study of two mission scenarios that aim to address the waste fuel leftover from missions stationed at SEL2. Both designs demonstrate that extremely low-cost asteroid flybys can be achieved in full even with additional relative arrival velocity and Time of Flight restrictions placed on top, largely in part due to the invariant manifold dynamics that can be harnessed in the CR3BP domain. Across both scenarios, 91 preliminary candidates were returned. 71 of

Table 8. Summary of final candidates across both Mission Scenarios post optimisation, ordered by Abs. Mag.

MS	Name	Abs. Mag.	TOF_{tot} (Mo.)	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})	MS	Name	Abs. Mag.	TOF_{tot} (Mo.)	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})
1	2004 FM17	19.36	16.74	0.1500	5.3282	1	2014 OV3	23.20	14.09	0.1500	4.5634
1	2004 DK1	21.30	18.63	0.1500	5.4206	2	2015 RF36	23.40	15.25	0.2083	4.2703
1	2014 OA2	21.48	20.22	0.1500	4.9677	2	2022 QM6	23.74	9.85*	0.2567	5.7791
1	2015 DP155	21.62	21.31	0.1499	4.2687	1	2008 MQ1	24.20	15.04	0.1499	4.3766
1	2009 CO5	21.85	26.59	0.1499	4.9666	2	2016 HO3	24.33	10.72*	0.2567	5.1457
1	2013 XV8	21.89	12.54	0.1500	3.3616	2	2021 RS	24.94	13.77	0.1300	2.441
1	2005 TF45	23.10	12.74	0.1500	3.8592	1	2024 CN5	24.98	14.32	0.1500	4.2470

Highlighted values are those that met the mission constraints. *TOF highlighted to call attention to closeness to a TOF of 9 months.

these objects were returned from MS 1, which used a manifold transfer to the Aphelion and a single burn for the final rendezvous. MS 1 proved to fall short of the TOF requirements even post optimisation, but overall houses a better asteroid search yield and optimisation outputs. MS 2 has shown to improve upon MS 1 by more than halving the manifold transfer duration but positions itself for lower asteroid search yield, and its optimisation results suggest further investigation at the EPF could produce trajectories with lower fuel cost. Overall, 14 objects have been shown to meet at least 1 of the 3 mission constraints within each of their trajectories. The most favourable candidate is likely to be 2013 XV8, and other objects of interest include 2004 FM17, 2022 QM6, 2021 RS, and 2016 HO3. Alongside these studies, an exact geometrical approach to calculating the MOID and a 2-stage hyperbolic EPF analysis to transfer directly to a NEO have been presented. Using the designs presented in this study not only are low-cost asteroid flybys achievable for the delta-v supplied - and at least another additional constraint - but they can be completed for significantly less delta-v margin if desired.

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